

Levantando Redis:

```
Last login: Wed Sep 30 21:38:54 on ttys000
tomy@MacBook-Neo-de-Tomy ~ % cd fixture2030-redis
cd: string not in pwd: fixture
tomy@MacBook-Neo-de-Tomy ~ % cd fixture2030-redis
tomy@MacBook-Neo-de-Tomy fixture2030-redis % ls
docs      README.md
docker-compose.yml  Makefile  scripts
tomy@MacBook-Neo-de-Tomy fixture2030-redis % docker compose up -d
unable to get image 'redis:latest': failed to connect to the docker API at unix:///Users/tomy/.docker/run/docker.sock; check if the path is correct and if the daemon is running: dial unix /Users/tomy/.docker/run/docker.sock: connect: no such file or directory
tomy@MacBook-Neo-de-Tomy fixture2030-redis % docker compose up -d
[+] up 2/2
[+] Network fixture2030-redis fixture2030-redis-net Created      0.0s
[+] Container fixture2030-redis Started      0.3s
tomy@MacBook-Neo-de-Tomy fixture2030-redis % docker compose ps
NAME                IMAGE                COMMAND                  SERVICE    CREATED      STATUS      PORTS
fixture2030-redis   redis:latest         "docker-entrypoint.s..." redis       2 minutes ago Up 2 minutes (healthy) 0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
```

```
Last login: Fri Oct 2 10:16:55 on ttys000
tomy@MacBook-Neo-de-Tomy ~ % docker version
Client:
 Version:           29.6.2
 API version:       1.55
 Go version:        go1.26.5
 Git commit:        dfc4efb
 Built:             Thu Jul 16 16:11:17 2026
 OS/Arch:           darwin/arm64
 Context:           desktop-linux

Server: Docker Desktop 4.83.0 (234302)
Engine:
 Version:           29.6.2
 API version:       1.55 (minimum version 1.40)
 Go version:        go1.26.5
 Git commit:        3d80467
 Built:             Thu Jul 16 16:13:03 2026
 OS/Arch:           linux/arm64
 Experimental:      false
containerd:
 Version:           v2.2.5
 GitCommit:         e53c7c1516c3b2bff98eb76f1f4117477e6f4e66
runc:
 Version:           1.3.6
 GitCommit:         v1.3.6-0-g491b69ba
docker-init:
 Version:           0.19.0
 GitCommit:         de40ad0
tomy@MacBook-Neo-de-Tomy ~ % docker compose version
Docker Compose version v5.3.1
tomy@MacBook-Neo-de-Tomy ~ %
```

```
tomy@MacBook-Neo-de-Tomy fixture2030-redis % docker compose exec -T redis redis-cli PING
PONG
tomy@MacBook-Neo-de-Tomy fixture2030-redis % docker compose exec -T redis redis-cli INFO server | grep -E 'redis_version|redis_mode|uptime_in_seconds'
redis_version:8.10.2
redis_mode:standalone
uptime_in_seconds:464
tomy@MacBook-Neo-de-Tomy fixture2030-redis %
```

Configuración de memoria:

```
tomy@MacBook-Neo-de-Tomy fixture2030-redis % docker compose exec -T redis redis-cli CONFIG GET maxmemory
maxmemory
134217728
tomy@MacBook-Neo-de-Tomy fixture2030-redis % docker compose exec -T redis redis-cli CONFIG GET maxmemory-policy
maxmemory-policy
volatile-lru
tomy@MacBook-Neo-de-Tomy fixture2030-redis %
```

Carga inicial:

```
[tomy@MacBook-Neo-de-Tomy fixture2030-redis % docker compose exec -T redis redis-cli < scripts/carga_muestra.redis ]
0
0
0
ERR unknown command '#', with args beginning with: 'Usuarios' 'sintéticos' 'reutilizados' 'de' 'Hito' '5.'

OK
OK
OK
OK
OK
ERR unknown command '#', with args beginning with: 'Contador' 'concurrente:' 'temporal' 'y' 'reconstruible.'

OK
ERR unknown command '#', with args beginning with: 'Ranking' 'temporal' 'de' 'figura' 'del' 'partido.'

1
1
1
1
1
ERR unknown command '#', with args beginning with: 'Una' 'actualización' 'de' 'voto/elección' 'es' 'atómica' 'sobre' 'el' 'score.'

13
=== Estado inicial cargado ===
0
3600
J-ARG-1
13
J-BRA-1
9
J-ARG-2
7
600
tomy@MacBook-Neo-de-Tomy fixture2030-redis %
```

Prueba de sesiones, renovación TTL, expiración por inactividad:

```
[tomy@MacBook-Neo-de-Tomy fixture2030-redis % grep -v '^[:space:]*#' scripts/sesiones.redis | docker compose exec -T redis redis-cli]
1
OK
QUEUED
QUEUED
8
1
=== Sesión creada ===
session_id
S-DEMO-001
usuario_id
U001
rol
fan
estado_acceso
ACTIVA
creado_en
2030-06-14T10:00:00-03:00
ultimo_acceso
2030-06-14T10:00:00-03:00
acciones
0
token_ref
DEMO-NO-SECRETO-001
1000
=== Recuperacion de sesion ===
U001
ACTIVA
Renovación atómica tras actividad válida
OK
QUEUED
QUEUED
QUEUED
1
1
0
1
session_id
S-DEMO-001
usuario_id
U001
rol
fan
estado_acceso
ACTIVA
creado_en
2030-06-14T10:00:00-03:00
ultimo_acceso
2030-06-14T10:05:00-03:00
acciones
1
token_ref
DEMO-NO-SECRETO-001
1000
0
OK
```

```

QUEUED
QUEUED
8
1
Estado previo al cierre
session_id
S-DEMO-002
usuario_id
U002
rol
fan
estado_acceso
ACTIVA
creado_en
2030-06-14T10:10:00-03:00
ultimo_acceso
2030-06-14T10:10:00-03:00
acciones
0
token_ref
DEMO-NO-SECRETO-002
1800
Finalización explícita de U002
0
session_id
S-DEMO-002
usuario_id
U002
rol
fan
estado_acceso
CERRADA
creado_en
2030-06-14T10:10:00-03:00
ultimo_acceso
2030-06-14T10:10:00-03:00
acciones
0
token_ref
DEMO-NO-SECRETO-002
OK
QUEUED
1
-2
0
6
1
TTL inicial de U003
2
1
Consecuencia de una sesión expirada
-2

```

Prueba de cache:

```

tomy@MacBook-Neo-de-Tomy fixture2030-redis % docker compose exec -T redis redis-cli < scripts/cache.redis
0
=== RF6/RF7: cache miss ===

ERR unknown command '#', with args beginning with: 'Simulacion' 'del' 'resultado' 'que' 'la' 'aplicacion' 'obtuvo' 'de' 'MongoDB.'

ERR unknown command '#', with args beginning with: 'TTL' '30' 's:' 'limite' 'de' 'permanencia' 'de' 'la' 'copia' 'si' 'no' 'ocurre' 'antes' 'una' 'invalidación' 'de' 'negocio.'

OK
=== Cache hit ===
{"partido_id":"P001","local":"Argentina","visitante":"Brasil","estado":"FINALIZADO","goles_local":2,"goles_visitante":1,"revision_fuente":1}
30
ERR unknown command '#', with args beginning with: 'Cambio' 'en' 'la' 'fuente' 'de' 'verdad' 'Cambio'

ERR unknown command '#', with args beginning with: 'En' 'la' 'arquitectura' 'real,' 'la' 'aplicación' 'actualiza' 'MongoDB' 'primero.'

ERR unknown command '#', with args beginning with: 'Después' 'invalida' 'la' 'copia' 'anterior' 'para' 'impedir' 'servir' 'datos' 'obsoletos.'

=== RF7: invalidación ante cambio de negocio ===
1

ERR unknown command '#', with args beginning with: 'Proximo' 'miss:' 'reconstrucción' 'de' 'la' 'copia' 'reconstrucción'

OK
=== Caché reconstruida desde la fuente de verdad ===
{"partido_id":"P001","local":"Argentina","visitante":"Brasil","estado":"FINALIZADO","goles_local":2,"goles_visitante":1,"revision_fuente":2}
30
=== RF7: clave ausente / comportamiento esperado ===
La aplicación debe consultar MongoDB y volver a poblar Redis; si Redis no está disponible, debe continuar desde la fuente de verdad sin usar Redis como autoridad.
tomy@MacBook-Neo-de-Tomy fixture2030-redis %

```

Prueba del ranking:

```
tomy@MacBook-Neo-de-Tomy fixture2030-redis % docker compose exec -T redis redis-cli ZREVRANGE f2030:ranking:partido:P001:figura 0 2 WITHSCORES
J-ARG-1
13
J-BRA-1
9
J-ARG-2
7
tomy@MacBook-Neo-de-Tomy fixture2030-redis %
```

Prueba de concurrencia:

```
tomy@MacBook-Neo-de-Tomy fixture2030-redis % bash scripts/prueba_concurrencia.sh
=== Prueba concurrente RF8 ===
Workers: 10
Incrementos por worker: 100
Esperado: 1000
Observado: 1000
TTL restante: 3598s
Duración observada: 716 ms
Resultado: OK - no se perdieron incrementos

Interpretación:
Los 10 workers ejecutan INCRBY sobre la misma clave en paralelo. INCRBY es una operación atómica dentro de Redis y evita el patrón GET-modificar-SET.
tomy@MacBook-Neo-de-Tomy fixture2030-redis %
```

Métricas:

```
tomy@MacBook-Neo-de-Tomy fixture2030-redis % docker compose exec -T redis redis-cli < scripts/metricas.redis
=== Version ===
# Server
redis_version:8.10.2
redis_git_sha1:00000000
redis_git_dirty:1
redis_build_id:de5ed0cfd99e0b21
redis_mode:standalone
os:Linux 6.12.76-linuxkit aarch64
arch_bits:64
monotonic_clock:ARM CNTVCT @ 24 ticks/us
multiplexing_api:epoll
atomicvar_api:c11-builtin
gcc_version:14.2.0
process_id:1
process_supervised:no
run_id:e2ad38083bd1c8dbd27a8f58a7b820a3b92011fe
tcp_port:6379
server_time_usec:1790948480477964
uptime_in_seconds:1212
uptime_in_days:0
hz:10
configured_hz:10
lru_clock:12563584
executable:/data/redis-server
config_file:
io_threads_active:0
listener0:name=tcp,bind=*,bind=::*,port=6379
=== Estadísticas de acceso: hits / misses / evictions ===
# Stats
total_connections_received:1252
total_commands_processed:1422
instantaneous_ops_per_sec:0
total_net_input_bytes:81178
total_net_output_bytes:17083
total_net_repl_input_bytes:0
total_net_repl_output_bytes:0
instantaneous_input_kbps:0.01
instantaneous_output_kbps:0.00
instantaneous_input_repl_kbps:0.00
instantaneous_output_repl_kbps:0.00
rejected_connections:0
sync_full:0
sync_partial_ok:0
sync_partial_err:0
expired_subkeys:0
expired_subkeys_active:0
expired_keys:2
expired_keys_active:2
expired_stale_perc:0.00
expired_time_cap_reached_count:0
expire_cycle_cpu_milliseconds:130
evicted_keys:0
```

```
=== Memoria ===  
# Memory  
used_memory:1900840  
used_memory_human:1.81M  
used_memory_rss:28790784  
used_memory_rss_human:27.46M  
used_memory_peak:2063544  
used_memory_peak_human:1.97M  
used_memory_peak_time:1790948456  
used_memory_peak_perc:92.12%  
used_memory_overhead:1484464  
used_memory_startup:1414760  
used_memory_dataset:416376  
used_memory_dataset_perc:85.66%  
allocator_allocated:12834536  
allocator_active:22020096  
allocator_resident:24117248  
allocator_muzzy:0  
total_system_memory:4108914688  
total_system_memory_human:3.83G  
used_memory_lua:32768  
used_memory_vm_eval:32768  
used_memory_lua_human:32.00K  
used_memory_scripts_eval:0  
number_of_cached_scripts:0  
number_of_functions:0  
number_of_libraries:0  
used_memory_vm_functions:34816  
used_memory_vm_total:67584  
used_memory_vm_total_human:66.00K  
used_memory_functions:192  
used_memory_hash_templates:72  
used_memory_scripts:192  
used_memory_scripts_human:192B  
maxmemory:134217728  
maxmemory_human:128.00M
```

```
=== Keyspace ===  
# Keyspace  
db0:keys=7,expires=7,avg_ttl=61975993,subexpiry=0  
=== Límite y política ===  
maxmemory  
134217728  
maxmemory-policy  
volatile-lru  
=== TTL de ejemplos ===  
1481  
-2  
-2  
3574  
-2  
=== Tipos ===  
hash  
none  
none  
string  
=== Inspección segura por cursor ===  
0  
f2030:sesion:U001  
f2030:demo:usuario:2  
f2030:demo:usuario:3  
f2030:demo:usuario:5  
f2030:demo:usuario:4  
f2030:contador:partido:P001:visitas  
f2030:demo:usuario:1  
tomy@MacBook-Neo-de-Tomy fixture2030-redis %
```

Medición:

```
tomy@MacBook-Neo-de-Tomy fixture2030-redis % bash scripts/medir.sh
Hito 7 - Medicion Redis
Fecha: 2026-10-02 10:42:38 -0300

--- Estado del contenedor ---
NAME          IMAGE          COMMAND          SERVICE    CREATED      STATUS      PORTS
fixture2030-redis  redis:latest  "docker-entrypoint.s..."  redis      21 minutes ago  Up 21 minutes (healthy)  0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp

--- Versión Redis ---
redis_version:8.10.2
redis_mode:standalone
uptime_in_seconds:1291

--- Configuración de memoria ---
maxmemory
134217728
maxmemory-policy
volatile-lru

--- redis-benchmark (GET/SET/INCR) ---
Metodo: 5000 requests por comando, 20 clientes, sobre el nodo local del contenedor.
SET: 200000.00 requests per second, p50=0.079 msec
GET: 312500.00 requests per second, p50=0.039 msec
INCR: 217391.30 requests per second, p50=0.079 msec

--- Microprueba de hit/miss ---
Hits de la microprueba: 20
Misses de la microprueba: 5
Tasa hit observada: 80,00%

--- Memoria observada ---
used_memory_human:1.88M
used_memory_peak_human:2.21M
maxmemory_human:128.00M

--- Métricas completas de stats relevantes ---
expired_keys:2
expired_keys_active:2
evicted_keys:0
keyspace_hits:5065
keyspace_misses:21

Medicion guardada en: docs/evidencia/benchmark_20261002_104238.txt
tomy@MacBook-Neo-de-Tomy fixture2030-redis %
```

Generar automaticamente la evidencia:

```
tomy@MacBook-Neo-de-Tomy fixture2030-redis % bash scripts/capturar_evidencia.sh
Hito 7 - Evidencia tecnica Redis
Fecha de ejecucion: 2026-10-02 10:43:28 -0300

===== RF1 / RNF1 / RNF2: ambiente =====
NAME          IMAGE          COMMAND          SERVICE    CREATED      STATUS      PORTS
fixture2030-redis  redis:latest  "docker-entrypoint.s..."  redis      22 minutes ago  Up 22 minutes (healthy)  0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
PONG
redis_version:8.10.2
redis_mode:standalone
uptime_in_seconds:1340
rdb_last_save_time:1798948559
aof_enabled:1
aof_rewrite_in_progress:0
maxmemory
134217728
maxmemory-policy
volatile-lru

===== Scripts de demostración =====

--- scripts/inicializacion.redis ---
RF1: disponibilidad de Redis
PONG
Version observada
# Server
redis_version:8.10.2
redis_git_sha1:00000000
redis_git_dirty:1
redis_build_id:de5ed0cfd99e0b21
redis_mode:standalone
os:linux 6.12.76-linuxkit aarch64
arch_bits:64
monotonic_clock:ARM CNTVCT @ 24 ticks/us
multiplexing_api:epoll
atomicvar_api:c11-builtin
gcc_version:14.2.0
process_id:1
process_supervised:no
```

```
=== Inspeccion segura por cursor ===
0
f2030:contador:partido:P001:meta
f2030:cache:partido:P001:resumen
f2030:sesion:U001
f2030:demo:usuario:2
f2030:demo:usuario:3
f2030:ranking:partido:P001:figura
f2030:demo:usuario:5
f2030:demo:usuario:4
f2030:contador:partido:P001:visitas
f2030:demo:usuario:1

===== Prueba concurrente =====
=== Prueba concurrente RF8 ===
Workers: 10
Incrementos por worker: 100
Esperado: 1000
Observado: 1000
TTL restante: 3599s
Duración observada: 604 ms
Resultado: OK - no se perdieron incrementos

Interpretación:
Los 10 workers ejecutan INCRBY sobre la misma clave en paralelo. INCRBY es una operación atómica dentro de Redis y evita el patrón GET-modificar-SET.

===== Métricas finales =====
expired_keys:2
expired_keys_active:2
evicted_keys:0
keyspace_hits:5098
keyspace_misses:26
used_memory_human:1.88M
used_memory_peak_human:2.21M
maxmemory_human:128.00M

Evidencia guardada en: docs/evidencia/ejecucion_20261002_104328.txt
```